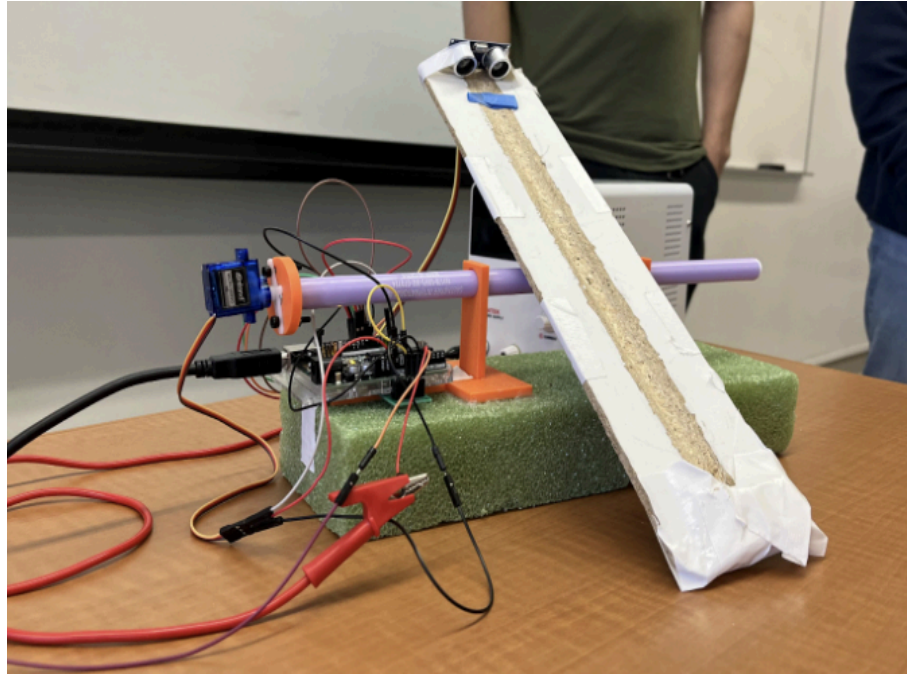


Eduardo Guerrero

• andrew_owner123@outlook.com

Portfolio of Projects

Tilt Control Mechanism



Goal:

Design and implement a control system with feedback to a lever arm in order to stabilize a moving spherical object and bring it to equilibrium.

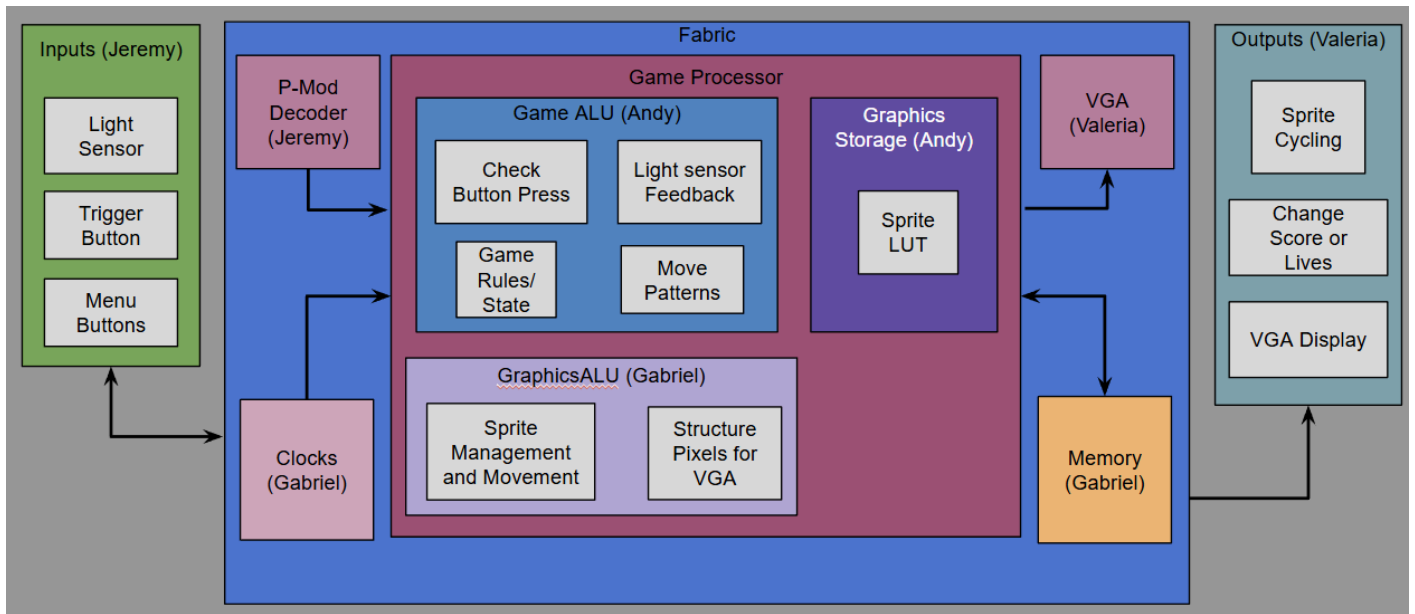
Challenges:

The main challenge was achieving stable control of the marble using real-time feedback. Small sensor fluctuations, mechanical misalignment, and the initial weight of the lever caused inconsistent behavior and oscillations. Additionally, tuning the PID proved difficult, since any aggressive gains caused instability while conservative gains resulted in a slow correction toward equilibrium.

Solution/Results:

To resolve these issues, the lever arm was sawed in half horizontally in order to reduce load on the servo motor. This helped to improve responsiveness. The ultrasonic sensor was adjusted and the PID controller was carefully tuned to balance stability and speed. With these changes the system was able to successfully stabilize the marble near the middle (equilibrium) and demonstrated the system's effective closed-loop feedback control despite minor mechanical limitations.

FPGA Based Game: Duck Hunt



Goal:

Design and implement an FPGA-based game system that recreates the core elements of Duck Hunt, including VGA graphics output, sprite rendering, game-state control, and hit registration of the target using a light sensor.

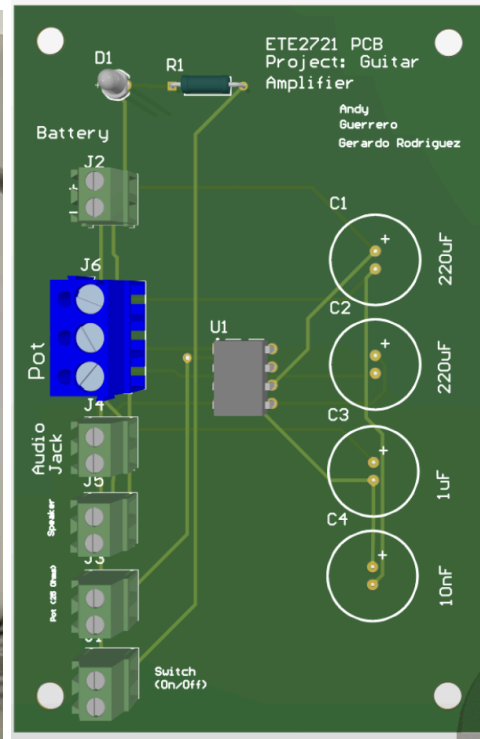
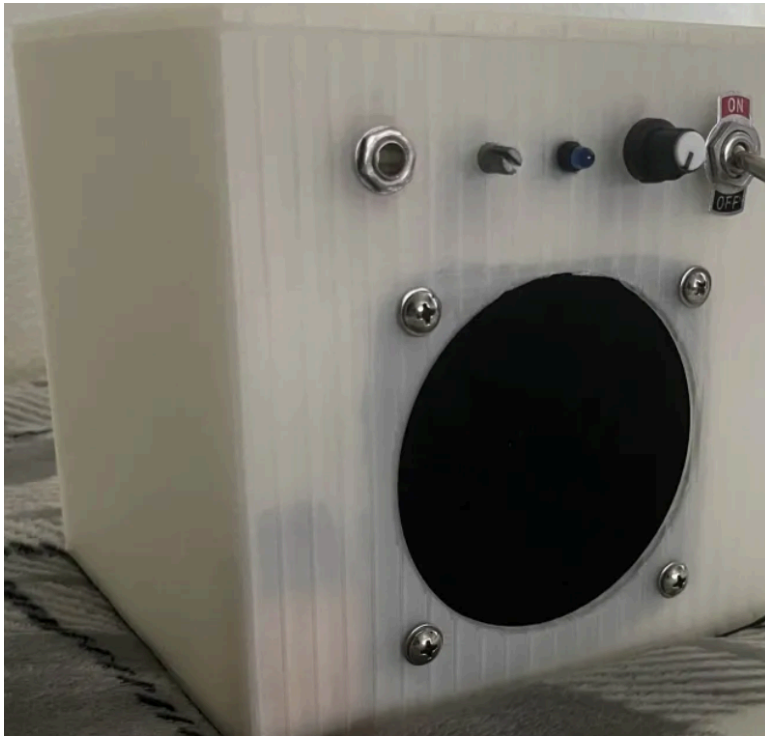
Challenges:

The main challenge was integrating multiple FPGA subsystems while maintaining proper timing and synchronization. Coordinating VGA signal generation, sprite rendering, game logic, and real-time light sensor input required careful clock management. Accurately detecting hits with the sensor on the moving block was another challenge we had to overcome.

Solution/Results:

These challenges were tackled by organizing the system into modular components, separating graphics, memory, input processing and game logic. LUTs were used for efficient sprite storage and a finite state machine was used to keep a smooth transition between the game states. The final system managed to successfully have the GAME state work, with a moving white block and a hit system that worked, turning on a purple LED on our Zybo whenever a hit that was successful was made.

Guitar Amplifier



Goal:

Design and manufacture a custom printed circuit board for a guitar amplifier with the use of Altium Designer, and integrate it into a 3D-printed enclosure. The amplifier was required to successfully amplify an electroacoustic guitar signal with adjustable gain and volume. We would also be demonstrating a proper PCB layout and enclosure integration.

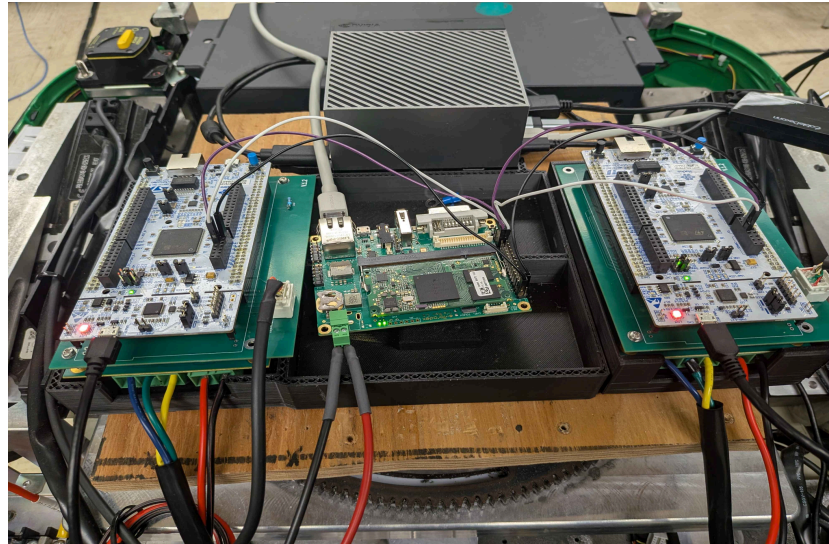
Challenges:

The primary challenges involved translating a working breadboard prototype into a compact and manufacturable PCB design. Finding appropriate component footprints in Altium Designer required substitutions and careful dimensional planning. Additionally, making sure that signals were routed correctly, reduction of noise, and correct component orientation (power polarity and LED placement) were critical to the avoidance of errors before fabrication.

Solution/Results:

These challenges were addressed with iterative schematic review, footprint substitution, and careful PCB layout optimization. Autorouting was used to make the connections quickly, and manual routing was utilized to correct the automated paths. Terminal blocks were added to simplify wiring and enclosure integration. A custom 3D enclosure was designed to securely house the PCB and the external components. The final manufactured PCB worked as intended, applying the guitar signal with adjustable gain and volume, showing a complete and functional end-to-end PCB design process.

Senior Project: Autonomous Localization Using ROS2



```
[outdoor_incremental_route_follower-15] [INFO] [1777354911.991213982] [outdoor_incremental_route_follower]: Sending waypoint 103/105: x=-2.22, y=-0.08, frame=map
[bt_navigator-10] [INFO] [1777354911.993541613] [bt_navigator]: Begin navigating from current location (0.31, -0.10) to (-2.22, -0.08)
[controller_server-6] [INFO] [1777354912.014588158] [controller_server]: Received a goal, begin computing control effort
[controller_server-6] [INFO] [1777354913.115277778] [controller_server]: Passing new path to controller.
[controller_server-6] [INFO] [1777354914.115205839] [controller_server]: Passing new path to controller.
[controller_server-6] [INFO] [1777354915.115188977] [controller_server]: Passing new path to controller.
[controller_server-6] [INFO] [1777354916.215197137] [controller_server]: Passing new path to controller.
[controller_server-6] [INFO] [1777354917.215192045] [controller_server]: Passing new path to controller.
[controller_server-6] [INFO] [1777354918.215192396] [controller_server]: Passing new path to controller.
[controller_server-6] [INFO] [1777354919.315204291] [controller_server]: Passing new path to controller.
[controller_server-6] [INFO] [1777354920.315196348] [controller_server]: Passing new path to controller.
[controller_server-6] [INFO] [1777354921.315201082] [controller_server]: Passing new path to controller.
[controller_server-6] [ERROR] [1777354922.015384988] [controller_server]: Failed to make progress
[controller_server-6] [WARN] [1777354922.015723344] [controller_server]: [follow_path] [ActionServer] Aborting handle.
[controller_server-6] [INFO] [1777354922.034218572] [local_costmap.local_costmap]: Received request to clear entirely the local_costmap
[controller_server-6] [INFO] [1777354922.036473428] [controller_server]: Received a goal, begin computing control effort
```

Goal:

The goal was to develop a ROS2-based localization and autonomous navigation system capable of determining a robot's position within a global environment through the fusion of GPS, IMU, and odometry data.

Challenges:

The primary challenge involves integrating multiple sensors and software systems into a stable localization framework using ROS2. GPS data alone was inconsistent outdoors due to signal drift and noise, while odometry data accumulated error over time.

Another major challenge was configuring ROS2 packages such as *robot_localization* and *Nav2* to properly communicate with the robot hardware and maintain an accurate position estimate in the global coordinate frame (the campus). Other challenges included debugging ROS2 nodes, coordinate frame transformations, and

maintaining reliable communication between the Jetson Orin, motor controllers, sensor systems, bumps while driving, and loss of connection while SSHing into the computer.

Solution/Results:

These challenges were addressed by the implementation of a sensor fusion programs using ROS2 and the *robot_localization* packages to combine GPS, odometry, and IMU data into a filtered localization estimate. Manual drive mode was developed to allow the robot to record GPS-based waypoint data while being controlled with a game controller.

This data was then used for autonomous route replaying through *Nav2*'s path planning and waypoint following. The project also included integration work with an NVIDIA Jetson Orin, STM32 Nucleo boards, and custom PCB improvements that reduced wiring complexity and improved hardware reliability. The final system successfully demonstrated outdoor localization, waypoint recording, and autonomous route-following behavior within a global coordinate system, without a base station needed.

The main goal of getting our robot to move autonomously was achieved and our future group will build upon this.